

Team 11 – MLB Offensive Power Analysis Final Report

Goals

Our business goal is to identify the strongest offensive teams for the 2025 MLB season. We aim to use this information to predict which teams are most likely to win the World Series. This analysis could be valuable for sports bettors, baseball analytics enthusiasts, or casual fans curious about potential winners.

Process

Our dataset contained player statistics from the 2023 and 2024 seasons. To transform individual player data into team-level insights, we aggregated player statistics and split the data into separate DataFrames for 2023 and 2024.

	Team	G	PA	AB	R	H	2B	3B	HR	RBI	...	CS	BB	SO	SH	SF	HBP	AVG	OBP	SLG	OPS
0	ARI	91.333333	365.888889	324.611111	913	1537	285	38	225	888	...	30	589	1336	33	65	89	0.258000	0.330500	0.424444	0.754944
1	ATL	103.000000	414.000000	372.470588	808	1506	306	17	234	763	...	18	593	1636	11	49	64	0.230706	0.302588	0.383412	0.686000
2	BAL	93.944444	378.833333	342.333333	835	1519	283	37	245	816	...	25	546	1543	6	48	63	0.240000	0.304500	0.403167	0.707667
3	BOS	78.157895	315.368421	284.157895	748	1367	299	30	192	696	...	42	493	1503	10	34	66	0.237000	0.310211	0.382211	0.692421

Based on our individual analysis, we determined OPS, RBI, AVG, SB, R, and H to be the most important statistics when it comes to determining how strong a player and or team is. However, some players have a low number of at-bats. Therefore, their AVG may be unrealistic and misleading, while their RBI, R, and H remained very low. To counteract this, for each team, we added their three best offensive players to their respective rows. This also allows us to factor “star power” into our analysis. Some teams might not be great all around, but they have a couple of superstars that bring them over the top.

We created a custom batting metric called *BAT_SCORE*. To calculate this, we determined each player’s fraction contribution to their team’s total R and RBI, highlighting their offensive impact. If a team scored a small number of runs in the season, but they have one player who scored most of them, this percentage will boost that player’s score. We added these percentages to OPS, a widely accepted MLB statistic combining how often a player gets on base (OBP) and how many bases he gets to per hit (SLG).

```
# Batting strength metric. Combines OPS (best avg metric for a batter's strength)
# and percentage of team runs and percentage of team RBIs
teams_and_players_24 = stats_24.merge(team_totals_24[["Team", "R", "RBI"]], on="Team")

teams_and_players_24["R_percent"] = teams_and_players_24["R_x"] / teams_and_players_24["R_y"]
teams_and_players_24["RBI_percent"] = teams_and_players_24["RBI_x"] / teams_and_players_24["RBI_y"]

teams_and_players_24["BAT_SCORE"] = teams_and_players_24["OPS"] + teams_and_players_24["R_percent"] + teams_and_players_24["RBI_percent"]

# Keep only the top three players from each team
teams_and_players_24 = teams_and_players_24.groupby("Team").apply(lambda x: x.nlargest(3, ["BAT_SCORE"])).reset_index(drop=True)
bat_score_24 = teams_and_players_24[["Team", "Player", "BAT_SCORE"]].sort_values(by="BAT_SCORE", ascending=False)
display(bat_score_24)
```

We then found the top three players on each team based on this offensive skill metric (*BAT_SCORE*). Then, we added columns to all teams for each of the three best players. We called them *P1_SCORE*, for the *BAT_SCORE* of the best player, *P2_SCORE* for the second best player, and *P3_SCORE* for the third best player, as seen below. To predict how good a team is going to be, however, we need a statistic for that.

	Team	Player	BAT_SCORE
54	NYN	Aaron Judge	1.478904
39	LAD	Shohei Ohtani	1.348448
84	TOR	Vladimir Guerrero	1.339943
33	KC	Bobby Witt	1.283430
55	NYN	Juan Soto	1.272442

	Team	G	...	AVG	OBP	SLG	OPS	P1_NAME	P1_SCORE	P2_NAME	P2_SCORE	P3_NAME	P3_SCORE
0	ARI	91.333333	...	0.258000	0.330500	0.424444	0.754944	Ketel Marte	1.140844	Joc Pederson	1.047980	Eugenio Suarez	1.000315
1	ATL	103.000000	...	0.230706	0.302588	0.383412	0.686000	Marcell Ozuna	1.179116	Matt Olson	1.014975	Jorge Soler	0.967840
2	BAL	93.944444	...	0.240000	0.304500	0.403167	0.707667	Gunnar Henderson	1.147062	Anthony Santander	1.047982	Colton Cowser	0.944774
3	BOS	78.157895	...	0.237000	0.310211	0.382211	0.692421	Rafael Devers	1.105563	Jarren Duran	1.090154	Tyler O'Neill	1.033574

To evaluate a team's overall success, we chose win-loss record as our target metric. If a team finishes at 0.600, meaning they won 60% of their games, they had a great season and are likely favorites to win going into the playoffs. However, if a team finishes at 0.250, meaning they only won 25% of their games, they are a bad team and will not make the playoffs, and should make drastic changes in the offseason.

Method

We trained a decision tree on this data. Decision trees are simple supervised models that are great at classification based on numerical data. They are easily interpretable, simple to construct, resistant to outliers, and provide additional information on feature importance. However, we needed to create a series of classes for our win-loss data because a decision tree cannot feasibly predict continuous values. To do this, we used the *pd.cut* function and binned the win-loss data into four bins. In baseball, an average of around 0.55 and above can usually get a team into the playoffs. These were:

- *Bad* (0 - 0.4)
- *Decent* (0.4 - 0.5),
- *Good* (0.5 - 0.57)
- *Great* (0.57 - 1)

Once all of the wrangling was complete, we combined the separate 2023 and 2024 datasets back into one. We chose H, AVG, SB, and the three player scores to be our features for the decision tree. This follows from our exploratory analysis. We split the data into test and train data at a split of 20/80. All of the input data is continuous, so we expect the tree might need to be complex; thus, we allowed a max depth of 7.

```
Accuracy: 0.5
Predictions: ['good' 'good' 'decent' 'good' 'good' 'decent' 'great' 'great' 'bad'
'great' 'decent' 'good']
Mean Squared Error: 0.75
```

Evaluation

We received an accuracy of 0.5 and an MSE of 0.75, which may not look great on paper, but we believe there are a couple of reasons for this. First, we are asking our decision tree to pick the exact bin a team *should* fall in based on their team, which is difficult given how many continuous variables there are. Second, our decision tree does not factor in a team's defense. Defense can make a huge difference in games. For example, a team may be great offensively, scoring 8 or 9 runs a game, but give up 10 runs. So, our decision tree would likely predict this team to be great, but their win-loss record would indicate otherwise. Finally, some teams get luckier than others or have more difficult schedules. A bad team might have a 0.500 record because it played a bunch of other bad teams. There are plenty of other factors as well, so 0.75 MSE is not bad in the grand scheme of things. We calculated the MSE because we wanted to see how far off our predictions were. If our decision tree took a team with a 0.6 win-loss record and classified it as "bad", that would be a problem. But if a decent team was classified as good, that's only a one bin difference.

Results

David created a Google Colab Notebook which, when given a team name, can webscrape its current batting roster. Then, we can take the output roster and run through some code to

```
def get_players(team):
    # The URL of the website to scrap
    roster_url = "https://www.mlb.com/" + team + "/roster/depth-chart"
    # gets HTML from the url
    soup = get_HTML(roster_url)
    # breaks HTML into a list of tables (where the players are stored)
    roster_tables = get_tables(soup)

    players = []
    for table in roster_tables:
        # Finds all the players from each table
        parse_table(table, players)

    return list(set(players))
```

produce a single row of this team with all of the same columns as the training data. This lets us use our decision tree on any current team we want to see how our model predicts they will perform offensively. We tested a few notable teams:

- Cincinnati Reds (David's favorite team) - *bad*
- Washington Nationals (Morgan's favorite team) - *decent*
- LA Dodgers (World Series Champs 2024) - *great*
- NY Yankees (Runner Up 2024) - *good*
- Philadelphia Phillies (ESPN Top 5 team) - *great*
- San Diego Padres (ESPN Top 5 Team) - *good*
- Colorado Rockies (ESPN Worst team) - *bad*
- Chicago Cubs (Best current offense) - *great*

Timeline

For our timeline, once we completed our individual exploratory data analysis, we planned our team analysis sketch. We quickly decided we were going to use a decision tree and knew we needed to find more data to properly train our model. David started by creating an offensive skill metric, Morgan found a way to find the top 3 players from each team, and John found the win-loss record from the previous seasons. These were simple tasks that we were able to complete in a few days. We reserved the rest of our time to create the decision tree. This took longer than anticipated, as there were many moving parts and ways to analyze. It put us behind on schedule to work on our interpretation and report, but we knew the decision tree would take up the bulk of time.

Future

This could be used in the future for sport analysts or sport bettors to continue to analyze baseball offensive statistics, whether it be for MLB or college. More years of data can be added to give more information for a stronger output. This is specifically for baseball, but could be manipulated for other sports to track other data. This could also be expanded to become more predictive and determine future statistics, and have higher accuracy.

Limitations

A limitation we realized was that we only had players' offensive statistics, and to find the team most likely to win the World Series, we were going to need more information. We were able to find the win-loss records from 2023 and 2024, so we could predict the team's skill level. Another limitation was that this only included offensive statistics. To have a more accurate analysis, we would also want to include defensive statistics.

Challenges

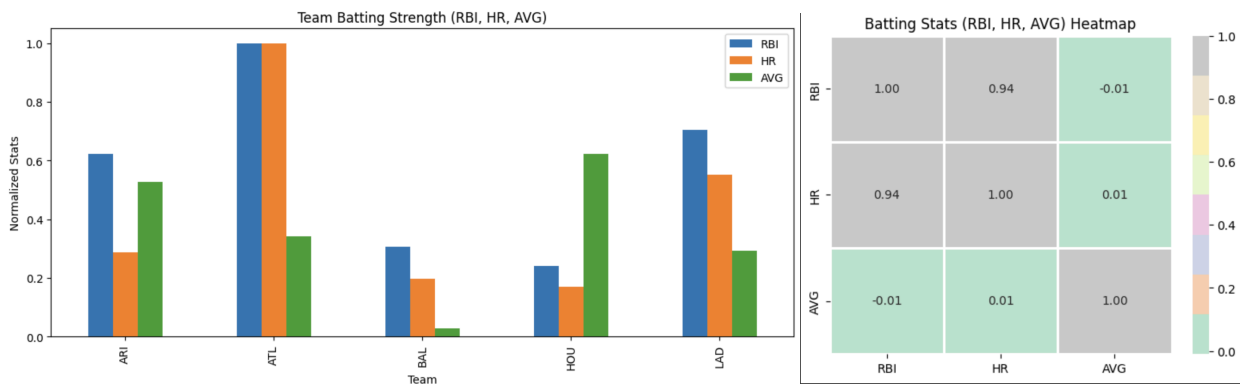
Some challenges faced were that the dataset had duplicate players since the data had entries for 2023 and 2024. This created an error when grouping by team to find the top players, so we split the data by 2023 and 2024 and found the top players from each team for each year. Another challenge was that the win-loss data was organized differently from the team data, and this needed to be altered so that they could contribute to the analysis.

Conclusion

In conclusion, we are confident in our model's efficacy, and we believe it reaches the goal of finding the best offensive teams. The model's predictions for some of this season's notable teams (as seen above) are fairly accurate so far. The Reds are struggling offensively, but are playing very well defensively and currently have a 0.500 record. The model predicted "bad" for them. The Yankees were predicted as "good" but they have been great offensively so far. The other teams are accurate based on how the season has been going. In the end, we believe we could confidently use this to find the most likely World Series champions, but if we want to improve its quality, we might consider including defensive statistics.

Individual Analysis – Morgan

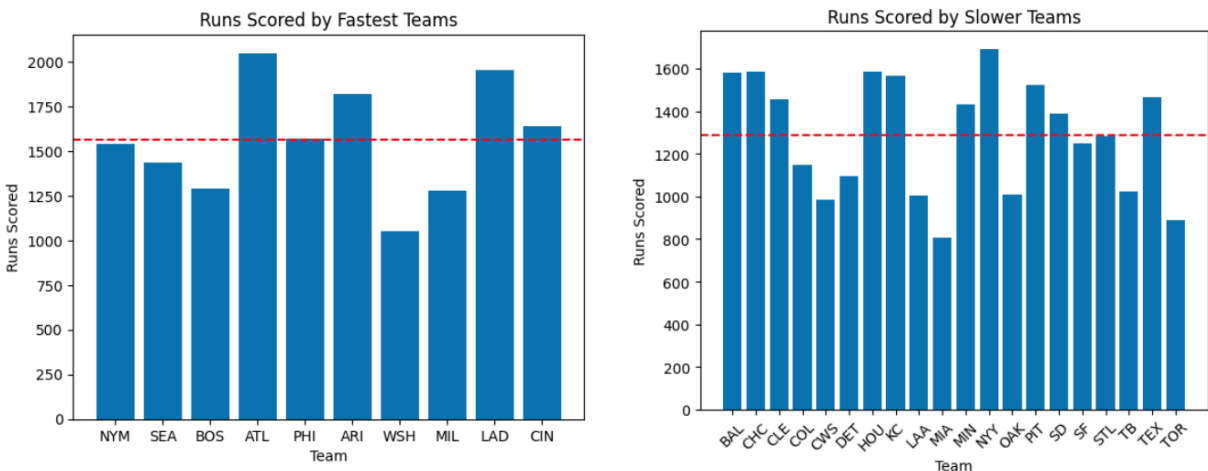
My goal was to find the best offensive MLB team by its scoring potential. I looked at Batting Average (AVG), how often a team gets on base and makes a hit, Home Runs (HR), which highlights a team's power-hitting ability, and Runs Batted In (RBI), an indicator of effectiveness in both home runs and other scoring plays. In my analysis, I found that AVG and RBI were two key indicators that could vary, but overall, they contributed to finding strong offensive teams.



For the team analysis, we used RBI to help determine the top players from each team, as this emphasizes the strong scoring players. We used AVG when making the decision tree to help divide teams into bins and rank their performance skills. In conclusion, I examined three statistics that showed player performance in the scoring space, and two of them aligned with our analysis and were integrated to help showcase the different statistics used to examine a team’s strength.

Individual Analysis – David

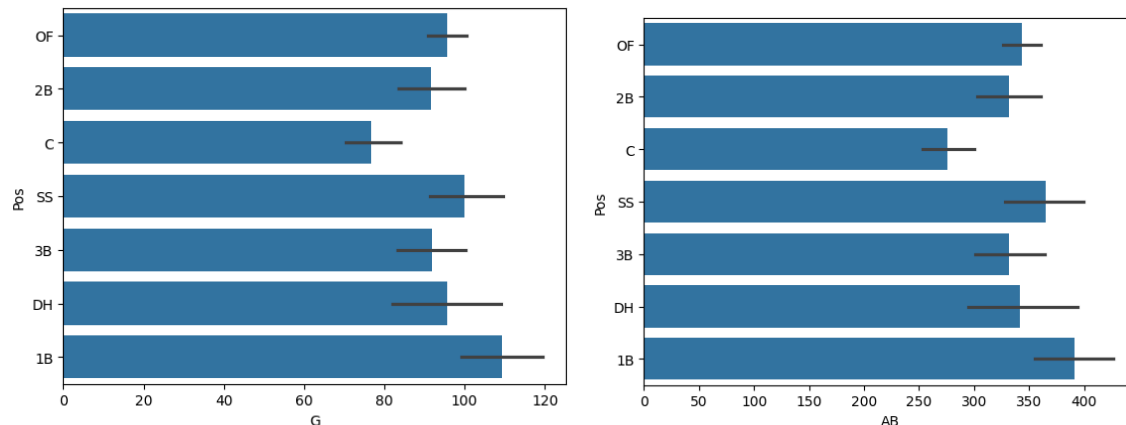
For my exploratory analysis, I analyzed how many stolen bases (SB) a team had, and each team’s on-base percentage (OBP), a measure of how often players get on base per time they go up to bat. These statistics are heavily tied to how fast the players on a team are. To steal a base, you must be quick. And, you are more likely to reach base if you are faster and can beat the throw. Based on these two statistics, I found the 10 teams with the most stolen bases and an above average OBP. I denoted these as the “fastest” teams and the rest being the “slower” teams. On average, the “fastest” teams scored significantly more runs than the “slower” teams, as seen by the dotted red lines.



However, speed is not the end all be all. The Washington Nationals scored just above 1,000 runs despite being a fast team. The Yankees scored upwards of 1,700 runs despite being in the “slower” teams group. But, based on this analysis, my team and I determined that stolen bases and OBP are important and should be considered when examining a team's offensive abilities. Therefore, we included stolen bases (SB) as a feature for our decision tree. Additionally, OBP is included within the OPS statistic, which we used to help find a team's most impactful players.

Individual Analysis – John

My goal was to find the best performing positions based on the amount of games played and number of at bats for each player, using positions, games played, and at bats which positions played the most amount of games and had the most amount of at bats. At bats and games are highly correlated and as shown with the two different plots with games, and positions, and at bats and positions. However because of the lack of a solid point of measurement to find a actual measurement of ability in the calculation such batting average the data I found has only a report of which positions had players that played the most which could but could be correlated but is not a direct indicator of performance. The business goal to find the best team could be affected by this individual goal, but it was not used and would likely not be impactful.



For the team analysis, I found the datasets for the win-loss-tie statistics for 2024, without this dataset prediction of team win-loss performance would not have been calculable, as batting statistics are not the only impactful data in baseball. The win-loss data was used to categorize the teams into four different categories based on overall performance, and was used by the decision tree model to create a model to classify teams based on performance.

Appendix

Resources

 mfrivera_scoring_eda_.pynb

 hw6_eda_david_sweasey

 relationship between games, position and at bat.ipynb

 data_analysis.ipynb

 team_getter.ipynb

[GitHub](#)

Data Dictionary

Variable	Description
Player	First and last name
Year	The year the data was collected from
Team	Team name abbreviation
Pos (Position)	Position abbreviation
G (Games)	Number of games played
AB (At bat)	Number of at bats
PA (Plate appearance)	Number of times player goes to the plate
R (Run)	Number of runs scored
H (Hit)	Number of hits
2B (2nd Base)	Number of double
3B (3rd Base)	Number of triples
HR (Home Run)	Number of home runs
RBI (Runs Batted In)	Number of runs scored from the player's plate appearance
SB (Stolen Bases)	The number of advances to a base that the player isn't allowed

CS (Caught Stealing)	Number of times a player is tagged while attempting to steal a base
AVG (Average)	Hits divided by total at-bats
OBP (On-base Average)	Number of times on base divided by the number of plate appearances
SLG (Slugging Percentage)	Total number of bases a player records per at-bat
OPS (On-base plus Slugging)	A combination of on-base percentage and slugging percentage
PX_SCORE	Player's batting strength metric using OPS, R, and RBI
PX_NAME	The name of the player associated with the strength metric